Monthly JBV / Research Policy Digest / JBV 与 RP 月度论文摘要

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Journal breakdown / 期刊分布: - Journal of Business Venturing / 创业研究期刊: 3 - Research Policy / 研究政策: 14

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Journal of Business Venturing / 创业研究期刊

Volume 42, Issue 1 | Published 2027-01

1. Between mission pull and governance burden: Geographic distance and deal realisation in impact investing

中文标题: 在使命拉动与治理负担之间:影响力投资中的地理距离与交易达成

Authors / 作者: Chukwuka Igboanua, Meike Siefkes, Simon J.D. Schillebeeckx, Henri Hakala

DOI: 10.1016/j.jbusvent.2026.106642 **Published / 发表日期**: 2027-01

Fields & Theories / 领域与理论:

- **Fields**: Entrepreneurial Finance, International Business, Social Entrepreneurship
- **领域**: 创业金融, 国际商务, 社会创业
- **Theories**: Transaction Cost Economics, Eclectic Paradigm (OLI), Location Logic
- **理论**: 交易成本经济学, 折衷范式 (OLI), 区位逻辑

Abstract / 摘要:

> Impact investing is widely expected to direct capital to locations where social and environmental challenges are most severe. Yet such opportunities are often found in places geographically distant from major pools of investment capital. We examine how geographic distance affects the realisation of impact investment deals and how host-country conditions alter that relationship.

Integrating transaction cost economics with the location logic of the eclectic paradigm, we argue that impact investment decisions reflect the joint consideration of anticipated transaction value and governance costs. Distance raises the costs of screening, contracting, monitoring, and supporting ventures, while severe sustainability challenges can increase a location's mission-specific attractiveness by signaling greater unmet need and potential scope for additionality. By contrast, trade and investment risk increases transaction hazards that make distant deals harder to execute. We test these arguments using PitchBook data on 3126 impact deals closed in 2022–2023. We analyse 3606 realised investor–venture ties against matched unrealised alternatives in a choice-based sampling design. Distance reduces the likelihood of deal realisation. Challenge severity modestly weakens the distance penalty for a subset of sampled dyads, whereas trade and investment risk strengthens it across most observations. These findings contribute to impact investing by explaining why severe unmet need does not automatically translate into realised investment relationships. They extend transaction cost reasoning in entrepreneurial finance by incorporating mission-adjusted transactional value. They refine international business location logic by distinguishing mission-specific locational attractiveness from host-country transaction hazard.

影响力投资被广泛预期会将资本引导至社会和环境挑战最严峻的地区。然而,此类机会往往位于地理上远离主要投资资本池的地方。我们研究了地理距离如何影响影响力投资交易的达成,以及东道国条件如何改变这种关系。结合交易成本经济学与折衷范式的区位逻辑,我们认为影响力投资决策反映了对预期交易价值和治理成本的共同考量。距离增加了筛选、签约、监督和支持创业企业的成本,而严峻的可持续发展挑战可以通过信号传递更大的未满足需求和潜在的额外性空间,从而增加特定地点的使命吸引力。相比之下,贸易和投资风险增加了交易危害,使得远程交易更难执行。我们利用 PitchBook 上 2022–2023 年完成的 3126 笔影响力交易数据对这些论点进行了测试。我们在基于选择的抽样设计中,分析了 3606 个已实现的投资者-创业企业关联,并与匹配的未实现替代方案进行了对比。距离降低了交易达成的可能性。挑战的严峻性适度削弱了部分样本对的距离惩罚,而贸易和投资风险则在大多数观察值中加强了这种惩罚。这些发现通过解释为什么严峻的未满足需求不会自动转化为已实现的投资关系,为影响力投资做出了贡献。它们通过纳入使命调整后的交易价值,扩展了创业金融中的交易成本推理。它们通过区分特定使命的区位吸引力与东道国交易危害,完善了国际商务的区位逻辑。

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2. Filling the legitimacy void: The value of sustainability orientation across investor segments in venture financing

中文标题: 填补合法性真空:创业融资中不同投资者群体对可持续发展导向的价值评估

Authors / 作者: Feilian Xia, James Thewissen, Paul P. Momtaz, Shuo Yan

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Fields & Theories / 领域与理论:

- **Fields**: Entrepreneurial Finance, Sustainable Entrepreneurship
- **领域**: 创业金融, 可持续创业
- **Theories**: Legitimacy Theory, Institutional Theory, Signaling Theory
- **理论**: 合法性理论, 制度理论, 信号理论

Abstract / 摘要:

When and why does sustainability attract entrepreneurial funding? Drawing on legitimacy theory, we argue that prior institutional backing segments ventures into distinct contexts in which public investors value sustainability orientation differently. Without such backing, sustainability can compensate for missing commercial validation by providing an alternative source of moral legitimacy. When institutional backing already certifies pragmatic legitimacy, sustainability may conflict with return-oriented expectations. Examining 2222 token-based crowdfunding campaigns, we find that sustainability orientation is positively associated with funding among non-backed ventures but negatively associated with funding among institutionally backed ventures. Consistent with this segmented valuation logic, non-backed ventures display stronger sustainability orientation. An LLM-based decomposition shows no statistically significant difference between the extent to which substantive and symbolic sustainability attenuate the funding disadvantage of non-backed ventures; their greater sustainability orientation is most clearly concentrated in symbolic social content. Our findings position sustainability not as a universally beneficial cue, but as a compensatory source of legitimacy whose value depends on what other types of legitimacy ventures already possess.

可持续发展何时以及为何能吸引创业资金?借鉴合法性理论,我们认为先前的机构支持将创业企业划分为不同的情境,在这些情境中,公众投资者对可持续发展导向的评价各不相同。在缺乏此类支持的情况下,可持续发展可以通过提供另一种道德合法性来源,来补偿

缺失的商业验证。当机构支持已经证明了务实合法性时,可持续发展可能与以回报为导向的预期发生冲突。通过对 2222 个基于代币的众筹项目进行研究,我们发现可持续发展导向与未获得机构支持的企业的融资呈正相关,但与获得机构支持的企业的融资呈负相关。与这种细分估值逻辑一致,未获得支持的企业表现出更强的可持续发展导向。基于大语言模型(LLM)的分解显示,实质性和象征性可持续发展在减轻未获得支持企业的融资劣势方面,其程度没有统计学上的显著差异;它们更强的可持续发展导向最明显地集中在象征性的社会内容上。我们的研究结果将可持续发展定位为一种补偿性的合法性来源,其价值取决于企业已经拥有的其他合法性类型,而非一种普遍有益的信号。

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Volume 41, Issue 6 | Published 2026-11

1. More than money: The symbolic role of stock options in tech worker attraction by technology startups

中文标题: 不仅仅是金钱:股票期权在技术创业公司吸引技术人员中的象征性作用

Authors / 作者: Di Tong, Haifeng Wang

DOI: 10.1016/j.jbusvent.2026.106640 **Published / 发表日期**: 2026-11

Fields & Theories / 领域与理论:

- **Fields**: Entrepreneurship, Human Resource Management, Organizational Behavior
- **领域**: 创业学, 人力资源管理, 组织行为学
- **Theories**: Signaling Theory, Symbolic Management, Sensemaking
- **理论**: 信号理论, 象征性管理, 意义建构

Abstract / 摘要:

The survival and growth of early-stage technology startups depend heavily on their ability to attract tech workers. It is popular for startups to offer stock options to rank-and-file tech workers, a practice rooted in the financial view that the financial prospects associated with these offerings help attract employees. Yet, recent research suggests tech workers often lack the knowledge necessary to properly evaluate stock options, casting doubt on this justification. To this end, we conducted a series of experiments to examine whether, and why, startup stock options may attract tech workers. We find a strong employee attraction effect for startups that offer stock options, which however is only partially driven by financial considerations. A more important driver appears to be tech workers' inference about a startup's unobservable qualities, particularly how it treats employees. While tech workers are aware of stock options' financial implications, they make financially calculated decisions only when explicitly prompted to do so. Recognizing that the financial view has limited applicability, our study highlights the importance of a symbolic approach to understanding stock options in startup recruiting. Our study contributes to the literature on startup recruiting and stock options by offering a more nuanced understanding of how tech workers make sense of stock option offerings and by highlighting the importance of a symbolic view alongside the prevailing financial view.

早期技术创业公司的生存和增长在很大程度上取决于其吸引技术人员的能力。创业公司向普通技术人员提供股票期权是很普遍的做法,这种做法根植于财务视角,即认为与这些期权相关的财务前景有助于吸引员工。然而,最近的研究表明,技术人员往往缺乏正确评估股票期权所需的知识,这使得上述理由存疑。为此,我们进行了一系列实验,以检验创业公司股票期权是否以及为何能吸引技术人员。我们发现,提供股票期权的创业公司具有强大的员工吸引效应,但这种效应仅部分由财务考虑驱动。一个更重要的驱动因素似乎是技术人员对创业公司不可观察特质的推断,特别是公司对待员工的方式。虽然技术人员意识到股票期权的财务影响,但他们只有在被明确提示时才会做出经过财务计算的决策。认识到财务视角的适用性有限,我们的研究强调了采用象征性视角来理解创业公司招聘中股票期权的重要性。通过对技术人员如何理解股票期权提供更细致的理解,并强调在盛行的财务视角之外象征性视角的重要性,本研究为创业招聘和股票期权的文献做出了贡献。

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Research Policy / 研究政策

Volume 55, Issue 10 | Published 2026-12

1. A cord of three strands: Industry-university-research collaboration and corporate ESG performance

中文标题: 三股合成的绳子:产学研合作与企业ESG表现

Authors / 作者: Wanyi Chen, Ying Wang, Hao Ren

DOI: 10.1016/j.respol.2026.105607 **Published / 发表日期**: 2026-12

Fields & Theories / 领域与理论:

- **Fields**: Corporate Sustainability, Innovation Management, Strategic Management
- **领域**: 企业可持续发展, 创新管理, 战略管理
- **Theories**: Boundary Spanning Theory, Triple Helix Model, Resource-Based View, Stakeholder Theory
- **理论**: 跨界理论, 三螺旋模型, 资源基础观, 利益相关者理论

Abstract / 摘要:

> Collaborative innovation equips firms with knowledge and resources aimed at competitiveness, but whether these also improve corporate environmental, social, and governance (ESG) performance is far from obvious. We examine industry-university-research (IUR) collaboration, that is, collaboration among industry, universities, and research institutes, a tripartite arrangement central to China's innovation system in which research institutes form a distinct third actor. Using a sample of Chinese A-share listed firms from 2009 to 2024, we find that IUR collaboration improves corporate ESG performance. The effect is economically meaningful: a one-standard-deviation increase in IUR is associated with an increase in ESG performance equivalent to 1.2% of its mean, and with increases of 4.8%, 2.1%, and 0.2% in the environmental, social, and governance dimensions, respectively. Adopting a boundary-spanning perspective, we show that the effect operates through three mechanisms, each tied to one ESG dimension: knowledge spillover that raises green awareness, resource integration that expands relational networks, and normative diffusion that strengthens human capital. The effect is stronger under high economic policy uncertainty, intense market competition, and advanced digital transformation, indicating that the value of boundary spanning is contingent on firms' external environment and internal capabilities. This study extends collaborative-innovation research from bilateral university-industry relationships to tripartite IUR collaboration in the Chinese context. It shifts the innovation-ESG literature from internally generated innovation to externally sourced, multi-party collaboration. It advances boundary-spanning theory by specifying when such collaboration improves sustainability performance. These findings offer guidance for policymakers and firms integrating collaborative innovation with sustainability goals.
>
> 协同创新为企业提供了旨在提升竞争力的知识和资源,但这些是否也能改善企业的环境、社会和治理(ESG)表现尚不明显。我们考察了产学研(IUR)合作,即工业界、大学和研究机构之间的三方协作安排,这是中国创新体系的核心,其中研究机构构成了独特的第三方主体。通过使用2009年至2024年中国A股上市公司的样本,我们发现产学研合作改善了企业的ESG表现。该效应在经济上具有显著意义:产学研合作每增加一个标准差,ESG表现就会增加其均值的1.2%,且环境、社会和治理维度的表现分别增加4.8%、2.1%和0.2%。从跨界视角出发,我们展示了该效应通过三个机制发挥作用,每个机制对应一个ESG维度:提高绿色意识的知识溢出、扩展关系网络的资源整合,以及强化人力资本的规范扩散。在经济政策不确定性高、市场竞争激烈以及数字化转型先进的情况下,这种效应更强,表明跨界的价值取决于企业的外部环境和内部能力。本研究将协同创新研究从双边的校企关系扩展到中国背景下的三方产学研合作。它将创新-ESG文献从内部产生的创新转向外部获取的多方协作。它通过明确此类合作何时能改善可持续发展表现,推进了跨界理论。这些发现为将协同创新与可持续发展目标相结合的政策制定者和企业提供了指导。

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2. A note on the changing appropriability conditions and technological opportunities for innovation in Japan; 1994–2020

中文标题: 关于日本创新获利性条件与技术机会变化的简报:1994–2020

Authors / 作者: Akiya Nagata, Akira Goto

DOI: 10.1016/j.respol.2026.105608 **Published / 发表日期**: 2026-12

Fields & Theories / 领域与理论:

- **Fields**: Innovation Economics, Strategic Management, National Innovation Systems
- **领域**: 创新经济学, 战略管理, 国家创新体系
- **Theories**: Appropriability Theory, Technological Opportunity, National Innovation Systems
- **理论**: 获利性理论, 技术机会, 国家创新体系

Abstract / 摘要:

> We report findings obtained by comparing two innovation surveys in Japan conducted in 1994 and 2020. A comparison of statistical means at two points in time, controlling for firm size and industry as covariates, reveals that the following three changes have occurred for appropriabilities and technological opportunities of innovation during the past quarter-century. First, the effectiveness of almost all the mechanisms by which firms' appropriate profits from innovation declined. Second, the imitation lag has lengthened considerably. Third, while universities and government research institutes have become more important sources of technological information, internal sources and competitors have become less important. These results suggest that Japan's innovation system has undergone a fundamental transformation between the two time points.
>
> 我们报告了通过比较1994年和2020年在日本进行的两次创新调查所获得的发现。在控制企业规模和行业作为协变量的情况下,对两个时间点的统计均值进行比较后发现,在过去的四分之一世纪里,创新的获利性和技术机会发生了以下三项变化。首先,企业从创新中获取利润的几乎所有机制的有效性都下降了。其次,模仿滞后时间显著延长。第三,虽然大学和政府研究机构已成为更重要的技术信息来源,但内部来源和竞争对手的重要性已经降低。这些结果表明,日本的创新体系在两个时间点之间发生了根本性的转型。

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3. Immigration and firm innovation: Evidence from Finnish manufacturing

中文标题: 移民与企业创新:来自芬兰制造业的证据

Authors / 作者: Terhi Maczulskij

DOI: 10.1016/j.respol.2026.105603 **Published / 发表日期**: 2026-12

Fields & Theories / 领域与理论:

- **Fields**: Economics of Innovation, Labor Economics, Innovation Management, Human Capital
- **领域**: 创新经济学, 劳动经济学, 创新管理, 人力资本
- **Theories**: Human Capital Theory, Knowledge Spillover Theory, Resource-Based View
- **理论**: 人力资本理论, 知识溢出理论, 资源基础观

Abstract / 摘要:

> This study examines the relationship between immigration and firm-level innovation in the Finnish manufacturing sector. The analysis exploits unique matched data linking employees' immigration status with firm-level patenting, process and product innovation activities, and innovation inputs. To address the potential endogeneity of firms' immigrant employment, a newly developed instrumental variables approach is applied, using the predicted historical geographic distribution of immigrants as an instrument. The results show that immigrant employment positively affects firms' patenting activity and the number of R&D workers, with these effects being statistically significant only among more productive firms. Furthermore, firms with a higher concentration of non-native employees exhibit stronger immigration-innovation effects, suggesting that the mechanisms linking immigration and innovation may depend on the scale and stage of immigration. Finally, the study finds that immigration enhances firm growth and boosts native employment.
>
> 本研究探讨了芬兰制造业中移民与企业层面创新之间的关系。分析利用了独特的匹配数据,将员工的移民身份与企业层面的专利申请、工艺和产品创新活动以及创新投入联系起来。为了解决企业雇佣移民可能存在的内生性问题,本研究采用了一种新开发的工具变量法,将预测的移民历史地理分布作为工具变量。结果表明,雇佣移民对企业的专利活动和研发人员数量有积极影响,且这些影响仅

> 在生产率较高的企业中具有统计显著性。此外,非本地员工集中度较高的企业表现出更强的移民-创新效应,这表明联系移民与创新的机制可能取决于移民的规模和阶段。最后,研究发现移民促进了企业增长并带动了本地员工就业。

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4. Stronger patent law increases the allocation of resources to external relative to internal R&D: Empirical evidence

中文标题: 强化专利法增加了外部研发相对于内部研发的资源分配:实证证据

Authors / 作者: Yun Hou, Xi Xiong, I.P.L. Png

DOI: 10.1016/j.respol.2026.105601 **Published / 发表日期**: 2026-12

Fields & Theories / 领域与理论:

- **Fields**: Innovation Management, Economics of Innovation, Intellectual Property Rights, Strategic Management
- **领域**: 创新管理, 创新经济学, 知识产权, 战略管理
- **Theories**: Property Rights Theory, Transaction Cost Economics, Resource Allocation Theory
- **理论**: 产权理论, 交易成本经济学, 资源分配理论

Abstract / 摘要:

> How should a technology firm adjust resource allocation between external and internal R&D in response to stronger patent protection? External R&D provides the firm with another channel of earnings to mitigate diminishing returns to internal R&D, but yields the firm only a fraction of the additional profit generated. Theoretically, if the marginal return to external R&D diminishes more slowly than the marginal return to internal R&D, the firm should increase external R&D proportionately more than internal R&D. Exploiting regional differences in the strengthening of patent protection due to the U.S. Court of Appeals for the Federal Circuit (CAFC), we find that the CAFC was associated with 35 percent more external R&D vis-à-vis 20 percent more internal R&D. The difference was more pronounced among firms more specialized in technology and in industries where patents were less effective.
>
> 面对更强的专利保护,技术型企业应如何调整外部研发与内部研发之间的资源分配?外部研发为企业提供了另一种收益渠道,以缓解内部研发收益递减的问题,但仅能为企业带来所产生额外利润的一小部分。从理论上讲,如果外部研发的边际收益递减速度慢于内部研发的边际收益递减速度,企业增加外部研发的比例应高于内部研发。利用美国联邦巡回上诉法院(CAFC)导致的专利保护加强在地区间的差异,我们发现 CAFC 与外部研发增加 35 percent 以及内部研发增加 20 percent 相关。这种差异在技术专业化程度较高的企业以及专利有效性较低的行业中更为显著。

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Volume 55, Issue 9 | Published 2026-11

1. Burning bridges or bridging divides: Geopolitics and collaboration in research and innovation

中文标题: 焚桥毁路还是弥合分歧:地缘政治与研发创新合作

Authors / 作者: Shreyas Gadgin Matha, Matte Hartog, Christian Chacua, Pierre-Alexandre Balland, Ricardo Hausmann, Muhammed Ali Yıldırım

DOI: 10.1016/j.respol.2026.105580 **Published / 发表日期**: 2026-11

Fields & Theories / 领域与理论:

- **Fields**: Science Policy, International Business, Geopolitics
- **领域**: 科学政策, 国际商务, 地缘政治

- **Theories**: Gravity Model of International Trade, Technology Sovereignty
- **理论**: 国际贸易引力模型, 技术主权

Abstract / 摘要:

> This paper analyzes the association between geopolitical alignment and international research and innovation collaborations using data spanning 2000–2023. Using ideal point distance measures derived from UN General Assembly voting patterns, we find that increases in geopolitical distance are associated with reductions in international collaborations. A one-unit increase in geopolitical distance (roughly the distance between a median country pair) over three years is associated with a 3.1% reduction in collaborations in scientific publications (0.8% per standard deviation). We also find that this relationship is asymmetric: worsening geopolitical relations are associated with reductions in collaborations in scientific publications that are roughly three times larger than the gains associated with equivalent improvements, suggesting that international research partnerships are easier to damage than to build. At the technology level, critical technologies exhibit approximately three times the expected level of international collaboration compared to other fields (conditional on country-pair effects), yet are also more sensitive to shifts in geopolitical alignment, raising concerns about technology sovereignty. At the same time, despite ongoing geopolitical tensions, overall collaboration patterns have not undergone substantial changes in recent years, suggesting resilience in international research and innovation networks.
>
> 本文利用 2000–2023 年的数据分析了地缘政治一致性与国际研发创新合作之间的关联。通过使用源自联合国大会投票模式的理想点距离测量,我们发现地缘政治距离的增加与国际合作的减少相关。三年内,地缘政治距离每增加一个单位(大约是中位数国家对之间的距离),科学出版物中的合作就会减少 3.1%(每标准差减少 0.8%)。我们还发现这种关系是不对称的:地缘政治关系的恶化与科学出版物合作的减少相关,其程度大约是同等程度改善所带来收益的三倍,这表明国际研究伙伴关系破坏容易建设难。在技术层面,关键技术表现出的国际合作预期水平大约是其他领域的三倍(受国家对效应约束),但对地缘政治一致性的变化也更为敏感,这引发了对技术主权的担忧。与此同时,尽管地缘政治紧张局势持续存在,但近年来的整体合作模式并未发生实质性变化,这表明国际研发创新网络具有韧性。

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2. Citations in 280 characters: Is social media presence worth it for elite journals?

中文标题: 280个字符中的引用:顶级期刊在社交媒体上的存在值得吗?

Authors / 作者: Philipp Lucas Wähler, Christos Kolympiris, Nicos Nicolaou

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Fields & Theories / 领域与理论:

- **Fields**: Sociology of Science, Scientometrics, Innovation Studies
- **领域**: 科学社会学, 科学计量学, 创新研究
- **Theories**: Matthew Effect, Status Theory, Information Diffusion
- **理论**: 马太效应, 地位理论, 信息扩散

Abstract / 摘要:

> Is it worthwhile for elite journals to actively promote their articles through an official social media account? We address this question by leveraging the staggered adoption of Twitter by three top economics journals: the Journal of Political Economy, the Quarterly Journal of Economics, and the Review of Economics and Statistics, to compare citation counts for promoted versus non-promoted articles. We find that institutional promotion increases citations, primarily by exposing research to audiences beyond economics and thereby fostering interdisciplinarity. The effect is larger for high-status authors, and we find suggestive evidence that it operates partly through authors' own networks. The way articles are presented, such as through visual aids, may further amplify their impact. These results show that active institutional promotion can enhance research dissemination even for journals that already command the lion's share of academic attention. At the same time, because the gains accrue disproportionately to already-visible authors, they raise concerns about inequities in scholarly recognition.
>
> 顶级期刊通过官方社交媒体账号积极推广其文章是否值得?我们利用三本顶级经济学期刊——《政治经济学杂志》(Journal of Political Economy)、《经济学季刊》(Quarterly Journal of Economics)和《经济学与统计学评论》(Review of Economics and Statistics)先后采用 Twitter 的时机,比较了被推广文章与未被推广文章的引用次数,从而探讨了这一问题。我们发现,机构推广增加了

> 引用量,这主要是通过将研究暴露给经济学以外的受众,从而促进了跨学科性。对于高地位作者,这种效应更大,我们还发现了暗示性证据,表明这种效应部分通过作者自身的人脉网络发挥作用。文章的呈现方式(如通过视觉辅助工具)可能会进一步放大其影响力。这些结果表明,即使是对于已经占据学术关注绝大部分份额的期刊,积极的机构推广也能增强研究的传播。与此同时,由于收益不成比例地流向了已经具有知名度的作者,这引发了对学术认可中不平等现象的担忧。

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3. Data and methods for identifying artificial intelligence-related patents

中文标题: 识别人工智能相关专利的数据与方法

Authors / 作者: Tianjun Wu, Chao Min, Waverly W. Ding, Guolong Wang, Kunpeng Zhang

DOI: 10.1016/j.respol.2026.105599 **Published / 发表日期**: 2026-11

Fields & Theories / 领域与理论:

- **Fields**: Innovation Studies, Scientometrics, Computational Social Science
- **领域**: 创新研究, 科学计量学, 计算社会科学
- **Theories**: Patent Analysis, Positive-Unlabeled Learning, Machine Learning
- **理论**: 专利分析, 正样本-无标签学习, 机器学习

Abstract / 摘要:

> This paper evaluates existing approaches to identifying artificial intelligence (AI)-related patents and introduces a novel, scalable framework for improving classification performance. Motivated by growing reliance on patent data in innovation research, we assess widely used methods, including patent class-based approaches and the USPTO's Artificial Intelligence Patent Dataset (AIPD), with an independent, human-expert-annotated ground-truth dataset. We document substantial performance limitations in existing approaches, particularly in terms of precision and generalizability. To address these challenges, we develop a CPC-informed label-refinement framework inspired by positive-unlabeled (PU) learning to construct high-quality training data. Our approach integrates hierarchical patent classification with data-driven refinement procedures to reduce label noise and improve representativeness. Using this refined dataset, we train a range of machine learning, deep learning, and transformer-based models. Our results show that models trained within our framework significantly outperform existing methods, including AIPD, achieving improvements over AIPD in F1 scores of approximately 18–21% on the same benchmark dataset. These gains are primarily driven by enhanced precision with only a modest reduction in recall, highlighting the central role of training data quality in classification performance. We further demonstrate the empirical value of improved AI patent identification through two applications, showing that the release of ChatGPT increased both the market valuation of AI patents and firms' allocation of innovative effort toward AI technologies. To support future research, we release our training data, source code, and a novel dataset of patent-level predictions (AIPat), with ongoing updates to reflect the evolving nature of AI innovation.
>
> 本文评估了现有的识别人工智能(AI)相关专利的方法,并引入了一个新颖的、可扩展的框架以提高分类性能。受创新研究中对专利数据日益增长的依赖所启发,我们利用一个独立的、由人类专家标注的基准数据集,评估了广泛使用的方法,包括基于专利分类的方法和美国专利商标局(USPTO)的人工智能专利数据集(AIPD)。我们记录了现有方法在性能上的重大局限,特别是在精确率和泛化性方面。为了应对这些挑战,我们开发了一个受正样本-无标签(PU)学习启发的基于CPC信息的标签细化框架,以构建高质量的训练数据。我们的方法将层级专利分类与数据驱动的细化程序相结合,以减少标签噪声并提高代表性。利用这一细化后的数据集,我们训练了一系列机器学习、深度学习和基于Transformer的模型。我们的结果显示,在我们的框架内训练的模型显著优于现有方法(包括AIPD),在同一基准数据集上,F1分数比AIPD提高了约18–21%。这些提升主要由精确率的增强驱动,而召回率仅有适度下降,突显了训练数据质量在分类性能中的核心作用。我们通过两个应用进一步展示了改进AI专利识别的经验价值,结果表明ChatGPT的发布既提高了AI专利的市场估值,也增加了企业向AI技术分配的创新努力。为了支持未来的研究,我们发布了训练数据、源代码以及一个新的专利级预测数据集(AIPat),并将持续更新以反映AI创新不断演变的本质。

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4. Farther apart, faltering partnerships? The impact of geographical separation on research collaboration

中文标题: 距离越远,伙伴关系越动摇?地理分离对研究合作的影响

Authors / 作者: Xin Liu, Ka Lok Pang, Yiwei Li, Zhiya Zuo, Dan Tian, Jiang Li

DOI: 10.1016/j.respol.2026.105604 **Published / 发表日期**: 2026-11

Fields & Theories / 领域与理论:

- **Fields**: Geography of Innovation, Economics of Science, Sociology of Science
- **领域**: 创新地理学, 科学经济学, 科学社会学
- **Theories**: Proximity Framework, Social Capital Theory
- **理论**: 邻近性框架, 社会资本理论

Abstract / 摘要:

> While technological advancements have reduced barriers to remote collaboration, the extent to which geographical proximity still shapes research outcomes remains a critical yet underexplored question. This study examines how geographical separation, defined as the loss of city-level co-location following collaborator mobility, affects the performance of established research collaborations. Using a quasi-experimental design, we analyze changes in research outcomes among 16,505 scientist pairs following inter-city relocation (treatment group) relative to intra-city relocation (control group). Our findings reveal that geographical separation leads to significant declines in both collaborative productivity (co-authored publications) and scientific impact (citation counts), underscoring the persistent role of proximity in sustaining effective collaboration. We further document important heterogeneity: interdisciplinary collaborations are more resilient in maintaining publication output but experience larger declines in scientific impact; same-stage collaborations are more vulnerable than cross-stage collaborations; and the negative effect is stronger in geographically larger countries. These findings highlight the enduring importance of geographical proximity in maintaining effective research collaboration.
>
> 虽然技术进步减少了远程合作的障碍,但地理邻近在多大程度上仍塑造着研究成果,仍然是一个关键但尚未得到充分探索的问题。本研究探讨了地理分离(定义为合作者流动后失去城市级别的同城办公状态)如何影响既有研究合作的表现。采用准实验设计,我们分析了 16,505 对科学家在跨城市搬迁(实验组)相对于同城搬迁(对照组)后研究成果的变化。我们的研究结果显示,地理分离导致合作生产力(合著论文)和科学影响力(引用次数)均显著下降,强调了邻近性在维持有效合作中的持久作用。我们进一步记录了重要的异质性:跨学科合作在维持论文产出方面更具韧性,但在科学影响力方面的下降幅度更大;同阶段合作比跨阶段合作更脆弱;且在地理面积较大的国家,这种负面影响更强。这些发现突显了地理邻近在维持有效研究合作中的持久重要性。

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5. Global market integration, national security salience, and new product development alliances in aircraft manufacturing

中文标题: 全球市场整合、国家安全显著性与飞机制造领域的新产品开发联盟

Authors / 作者: João Albino-Pimentel, Pierre Dussauge, Louis Mulotte, Elie Sung, Stephen Tallman

DOI: 10.1016/j.respol.2026.105583 **Published / 发表日期**: 2026-11

Fields & Theories / 领域与理论:

- **Fields**: Strategic Management, Innovation Management, International Business, Geopolitics
- **领域**: 战略管理, 创新管理, 国际商务, 地缘政治
- **Theories**: Resource-Based View, Resource Dependence Theory
- **理论**: 资源基础观, 资源依赖理论

Abstract / 摘要:

> We investigate how the global market integration of a firm's home country and the security salience of specific projects shape firms' propensity to form alliances during new product development. The Resource-Based View suggests that the development of technologically complex products often requires unique resources that may be unavailable domestically, driving firms to seek international partnerships. However, we propose that, in high-technology industries where national interests are at stake, alliances in the product development phase face unique constraints. They involve partner firms committing non-redeployable resources to a specific, well-defined product and to new assets and processes to manufacture it. Moreover, because this phase

> determines which specific features become available for commercial or military deployment, these alliances attract heightened scrutiny from governments, especially in security-sensitive projects. We thus propose that the well-established relationship between product complexity and alliance formation is negatively moderated by two major geopolitical factors: (i) the global market integration of the firm's home country and (ii) the national security salience of the project. Both conditions enable or enhance alternative pathways to resource access that reduce dependence on alliances. Using data on new aircraft development projects, we find support for our predictions. Our findings contribute to the understanding of how home-country global market integration and project national security salience shape firms' innovation strategies by affecting alliance formation decisions during product development.
>
> 我们研究了企业母国的全球市场整合程度以及特定项目的安全显著性如何影响企业在新产品开发过程中组建联盟的倾向。资源基础观(RBV)指出,技术复杂产品的开发通常需要国内无法获得的独特资源,从而驱动企业寻求国际合作伙伴。然而,我们提出,在涉及国家利益的高科技行业中,产品开发阶段的联盟面临着独特的约束。这些联盟涉及合作伙伴将不可转用的资源投入到特定的、定义明确的产品以及用于制造该产品的新资产和流程中。此外,由于这一阶段决定了哪些特定功能可用于商业或军事部署,这些联盟会引起政府的高度关注,尤其是在安全敏感的项目中。因此,我们提出,产品复杂性与联盟形成之间公认的关系受到两个主要地缘政治因素的负向调节:(i)企业母国的全球市场整合程度,以及(ii)项目的国家安全显著性。这两种情况都能够提供或增强获取资源的替代路径,从而减少对联盟的依赖。利用新飞机开发项目的数据,我们发现研究预测得到了支持。我们的研究结果有助于理解母国全球市场整合和项目国家安全显著性如何通过影响产品开发期间的联盟形成决策,进而塑造企业的创新战略。

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6. Minds and machines: Rethinking absorptive capacity in the age of artificial intelligence

中文标题: 心智与机器:人工智能时代的吸收能力再思考

Authors / 作者: Mattia Pedota

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Fields & Theories / 领域与理论:

- **Fields**: Innovation Management, Organizational Learning, Information Systems
- **领域**: 创新管理, 组织学习, 信息系统
- **Theories**: Absorptive Capacity, Human-AI Collaboration, Organizational Learning Theory
- **理论**: 吸收能力, 人机协作, 组织学习理论

Abstract / 摘要:

> The absorptive capacity construct derives its functions, dimensions, and internal relationships from the implicit assumption that humans are the only learning agents underlying knowledge absorption. I posit that the emergence of artificial intelligence (AI) challenges this assumption and requires a re-examination of the construct. Through a theory-building abductive study combining conceptual mapping with qualitative interviews, this work investigates the complex relationship between AI and knowledge absorption. It theorizes new dimensions of absorptive capacity pertaining to AI-driven processes and analyzes the interactions between these new dimensions and the traditional facets of the construct. Ultimately, the paper seeks to reaffirm the importance of absorptive capacity in the age of AI and develop a nuanced understanding of AI's role in reshaping the construct's antecedents, mechanisms, and implications.
>
> 吸收能力这一构念的功能、维度和内部关系源于一个隐含假设,即人类是知识吸收背后唯一的学习主体。我提出,人工智能(AI)的兴起挑战了这一假设,并要求对该构念进行重新审视。通过一项结合概念映射与定性访谈的理论构建式溯因研究,本工作调查了AI与知识吸收之间复杂的关系。它理论化了与AI驱动过程相关的吸收能力新维度,并分析了这些新维度与该构念传统层面之间的相互作用。最终,本文旨在重申吸收能力在AI时代的重要性,并对AI在重塑该构念的前因、机制和影响方面的作用形成细致的理解。

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7. Modeling innovation ecosystem dynamics through interacting reinforced Bernoulli processes

中文标题: 通过相互作用的强化伯努利过程模拟创新生态系统动态

Authors / 作者: Giacomo Aletti, Irene Crimaldi, Andrea Ghiglietti, Federico Nutarelli

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Fields & Theories / 领域与理论:

- **Fields**: Economics of Innovation, Innovation Studies, Scientometrics
- **领域**: 创新经济学, 创新研究, 科学计量学
- **Theories**: Cumulative Innovation, Knowledge Spillovers, Evolutionary Economics
- **理论**: 累积创新, 知识溢出, 演化经济学

Abstract / 摘要:

> Innovation is cumulative and interdependent: successful inventions build on prior knowledge within technological fields and may also affect success across related ones. Yet these dimensions are often studied separately in the innovation literature. This paper asks whether patent success across technological categories can be represented within a single dynamic framework that jointly captures within-category reinforcement, cross-category spillovers, and a set of aggregate regularities observed in patent data. To address this question, we propose a model of interacting reinforced Bernoulli processes in which the probability of success in a given category depends on past successes both within that category and across other categories. The framework yields joint predictions for success probabilities, cumulative successes, relative success shares, and cross-category dependence. We implement the model using granted US patent families from GLOBAL PATSTAT (1980–2018), defining category-specific success through a cohort-normalized forward-citation index. The empirical analysis shows that successful innovations continue to accumulate, but less than proportionally to the growth in patent opportunities, while technological categories remain interdependent without becoming homogeneous. Under a mean-field restriction, the model-based inferential exercise points to a positive but non-maximal interaction across technological categories.
>
> 创新具有累积性和相互依赖性:成功的发明建立在技术领域先前的知识基础之上,也可能影响相关领域的成功。然而,在创新文献中,这些维度通常被分开研究。本文探讨了技术类别的专利成功是否可以在一个单一的动态框架内表示,该框架共同捕捉了类别内的强化、跨类别的溢出效应以及在专利数据中观察到的一组总体规律。为了解决这个问题,我们提出了一个相互作用的强化伯努利过程模型,其中给定类别的成功概率取决于该类别内部以及其他类别的过去成功经验。该框架为成功概率、累积成功、相对成功份额和跨类别依赖性提供了联合预测。我们利用来自 GLOBAL PATSTAT(1980–2018)的已授权美国专利族实施了该模型,并通过队列归一化的前向引用指数定义了特定类别的成功。实证分析表明,成功的创新继续累积,但其增长比例低于专利机会的增长,同时技术类别保持相互依赖而未变得同质化。在平均场限制下,基于模型的推断练习指向技术类别之间存在积极但非最大程度的相互作用。

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8. R&D investment, technology adoption and productivity dynamics: An empirical study of the productivity dispersion in Canada

中文标题: 研发投入、技术采用与生产率动态:加拿大生产率离散度的实证研究

Authors / 作者: Jianmin Tang, Weimin Wang, Sadaf Yalinejad

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Fields & Theories / 领域与理论:

- **Fields**: Economics of Innovation, Industrial Organization, Productivity Analysis
- **领域**: 创新经济学, 产业组织, 生产率分析
- **Theories**: Technology Diffusion, Firm Heterogeneity, Endogenous Growth Theory
- **理论**: 技术扩散, 企业异质性, 内生增长理论

Abstract / 摘要:

> Firm productivity dispersion has widened across many OECD countries and is often interpreted as evidence of persistent divergence between frontier and laggard firms, reflecting weakened technology diffusion and growing structural rigidities. This study revisits that narrative using comprehensive Canadian microdata and shows that conclusions about productivity divergence depend on how frontier and laggard firms are defined. Consistent with the existing literature, annually selected frontier and

laggard firms exhibit a widening productivity gap over time. However, when the same firms are followed as fixed cohorts, the analysis reveals substantial productivity convergence rather than divergence. This contrast reflects the highly dynamic nature of firm productivity: approximately 15% of lagging firms move into the middle or frontier groups within three years, while 37% of frontier firms no longer remain at the frontier over the same period. The study demonstrates that the widely documented productivity gap between frontier and laggard firms partly reflects changes in firm composition rather than persistent differences among fixed groups of firms. The study further shows that sustained investments in R&D and technology are consistently associated with productivity mobility and the ability to maintain frontier status, with such investments being more prevalent among firms that improve their productivity performance or remain at the productivity frontier. These findings have important implications for interpreting productivity dispersion and for designing policies aimed at fostering productivity growth.

在许多经合组织(OECD)国家,企业生产率的离散度有所扩大,这通常被解释为前沿企业与落后企业之间存在持续分化的证据,反映了技术扩散的减弱和结构性僵化的增长。本研究利用详尽的加拿大微观数据重新审视了这一说法,并表明关于生产率分化的结论取决于如何定义前沿企业和落后企业。与现有文献一致,每年选出的前沿企业和落后企业随时间推移表现出生产率差距扩大的趋势。然而,当将相同的企业作为固定队列进行追踪时,分析显示出显著的生产率收敛而非分化。这种对比反映了企业生产率高度动态的本质:约15%的落后企业在三年内进入了中间组或前沿组,而37%的前沿企业在同一时期内不再保持前沿地位。研究表明,被广泛记录的前沿企业与落后企业之间的生产率差距,部分反映了企业构成的变化,而非固定企业群体之间持续存在的差异。研究进一步表明,持续的研发和技术投资与生产率流动性以及维持前沿地位的能力始终相关,此类投资在提高生产率表现或保持生产率前沿的企业中更为普遍。这些发现对于解释生产率离散度以及设计旨在促进生产率增长的政策具有重要意义。

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9. Risk and the discursive construction of quantum technologies in Australia: How geopolitical and economic threats are used in hyping emerging technologies

中文标题: 风险与澳大利亚量子技术的论述构建:地缘政治和经济威胁如何被用于炒作新兴技术

Authors / 作者: Gabriella Skoff, Jarryd Daymond, Steven Maguire

DOI: 10.1016/j.respol.2026.105582 **Published / 发表日期**: 2026-11

Fields & Theories / 领域与理论:

- **Fields**: Science and Technology Studies, Innovation Studies, Sociology of Technology
- **领域**: 科学技术研究, 创新研究, 技术社会学
- **Theories**: Sociology of Expectations, Risk Society
- **理论**: 预期社会学, 风险社会

Abstract / 摘要:

In this paper, we examine the role of hype in the discursive construction of emerging quantum technologies. Our findings reveal that, as these technologies transition from academia to industry, they are the object of 'hyping' in which the discourses of risk and responsibility figure prominently. We find that hype – a collective vision around which attention, excitement, and expectations escalate – emerges from a process of discursively constructing emerging technologies as risk management solutions to highly salient geopolitical and economic risks to state actors, while acknowledging that they may pose less salient, novel social risks to citizens and social groups within a state. We also find that the former construction overwhelms the latter, which has important implications for what actions associated with developing these emerging technologies are considered responsible. As a result, and building on the sociology of expectations, we develop a discursive model showing how hyping mobilizes risk and responsibility to shape the meanings attached to emerging technologies and, in turn, the expectations that form around them – with important implications for the direction and pace of technological development.

在本文中,我们考察了"炒作"(hype)在新兴量子技术的论述构建中的作用。我们的研究结果表明,随着这些技术从学术界转向工业界,它们成为了"炒作"的对象,其中风险和责任的论述占据了突出地位。我们发现,炒作——一种围绕其注意力、兴奋感和预期不断升级的集体愿景——源于一种论述构建过程,即通过将新兴技术构建为应对国家行为体面临的高度显著的地缘政治和经济风险的风险管理解决方案,同时承认它们可能给国家内的公民和社会群体带来显著性较低的、新颖的社会风险。我们还发现,前者的构建压倒了后者,这对于开发这些新兴技术相关的哪些行动被认为是负责任的具有重要意义。因此,基于预期社会学,我们开发了一个论述模型,展示了炒作如何动员风险和责任来塑造赋予新兴技术的含义,进而塑造围绕它们形成的预期——这对技术发展的方向和速度具有重要影响。

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10. Science under sanctions: The impact of the entity list on Chinese academic research

中文标题: 制裁下的科学:实体清单对中国学术研究的影响

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Fields & Theories / 领域与理论:

- **Fields**: Economics of Science, Science Policy, International Relations
- **领域**: 科学经济学, 科学政策, 国际关系
- **Theories**: Social Network Theory, Resource Dependence Theory
- **理论**: 社会网络理论, 资源依赖理论

Abstract / 摘要:

> This study examines the effects of geopolitical tensions and technology-related sanctions on academic research, with a focus on the U.S. Entity List's inclusion of Chinese academic institutions. Using a comprehensive dataset of academic publications from the Open Academic Graph, we employ a stacked event study design to estimate the causal effects of the Entity List sanctions on academic research productivity. Our findings reveal that Entity List sanctions reduce the quantity but improve the quality of research output among affected Chinese researchers. Mechanism analyses, supplemented by semi-structured interviews, show that these changes are driven by research diversification and collaboration network reconfiguration. Sanctioned researchers shift toward new topics and form partnerships with non-listed Chinese institutions, which serve as intermediaries that restore access to restricted resources and expose researchers to novel knowledge. Notably, we find no significant decline in collaboration with U.S.-based researchers, suggesting that sanctions reconfigure rather than sever international networks. Theoretically, this study extends the disruption-as-catalyst logic to geopolitically imposed resource constraints, showing that sanctions can trigger productive diversification. By demonstrating that sanctions affect research quantity and quality in opposing directions through specific behavioral mechanisms, this study also reconciles inconsistent findings in prior literature and advances our understanding of how geopolitical tensions reshape the global academic research landscape.
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> 本研究考察了地缘政治紧张局势和技术相关制裁对学术研究的影响,重点关注美国实体清单将中国学术机构列入其中的情况。利用来自 Open Academic Graph 的全面学术出版物数据集,我们采用堆叠事件研究设计来估计实体清单制裁对学术研究生产力的因果效应。我们的研究结果显示,实体清单制裁降低了受影响中国研究者的研究产出数量,但提高了产出质量。机制分析辅以半结构化访谈表明,这些变化是由研究多样化和合作网络重构驱动的。受制裁的研究者转向新课题,并与未列入清单的中国机构建立合作伙伴关系,这些机构充当了恢复受限资源获取渠道并使研究者接触到新颖知识的中介。值得注意的是,我们发现与美国研究者的合作并未显著下降,这表明制裁重构了而非切断了国际网络。在理论上,本研究将"干扰作为催化剂"的逻辑扩展到地缘政治施加的资源约束上,表明制裁可以触发生产性的多样化。通过证明制裁通过特定的行为机制对研究数量和质量产生相反方向的影响,本研究还调和了先前文献中不一致的发现,并推进了我们对地缘政治紧张局势如何重塑全球学术研究格局的理解。

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